

SymPy Bug Card

Bug 37: Wrong Sign in an Antiderivative on the Negative Real Branch

Task. Compute an antiderivative of the algebraic integrand

$$\int \frac{dx}{x\sqrt{x^2-1}}.$$

SymPy's answer. SymPy returns

$$\text{integrate}\left(\frac{1}{x\sqrt{x^2-1}}, x\right) = \begin{cases} i \operatorname{acosh}(1/x), & 1/|x^2| > 1, \\ -\operatorname{asin}(1/x), & \text{otherwise.} \end{cases}$$

Correct answer. On the negative real branch $x < -1$, the derivative of the returned antiderivative should be

$$\frac{1}{x\sqrt{x^2-1}}.$$

Explanation. At $x = -3$, the integrand is approximately -0.11785113019775792 , but differentiating SymPy's fallback branch $-\operatorname{asin}(1/x)$ gives $+0.11785113019775792$. The fallback derivative is $1/(|x|\sqrt{x^2-1})$, which is correct on $x > 1$ but has the opposite sign on $x < -1$.

Likely root cause. The diagnosis narrows the issue to the Meijer-G indefinite integration path in `sympy/integrals/meijerint.py`, where a rewrite, hyperexpansion, polar power denesting, and unpolarification sequence appears to collapse the $x > 1$ and $x < -1$ branches into one $-\operatorname{asin}(1/x)$ fallback. This is a new instance of a known failure mode; the same $\sqrt{x^2-1}$ integration family appears in public issues [sympy/sympy#20982](#) ("lost sign when $x < 0$ ") and [sympy/sympy#23566](#) (the $\operatorname{acosh}/\operatorname{asin}$ branch handling), but neither reports the exact $1/(x\sqrt{x^2-1})$ indefinite sign error.

Artifact (GitHub). [bug-report/bug-037-wrong-sign-in-an-antiderivative-on-the-negative-real-branch/](#)